Clay-CD-not-certified: A Lean 4 Illustration of Why Linear Enstrophy-History Growth Does Not Imply BKM Blowup

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/Clay-CD-not-certified
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Abstract

A short Lean 4 illustration (`lean/NotBlowup.lean`). With the vorticity supremum held at a constant M, the Beale–Kato–Majda integral $\int_0^T M\,dt = MT$ is finite on every finite interval, while the quadratic history $\int_0^T M^2\,dt = M^2T$ grows linearly in T. The two Lean statements, `bounded_M_keeps_bkm_finite` and `history_can_grow_while_bkm_stays_finite`, hold by definitional unfolding (`rfl`) of the model functions; they are arithmetic, not analysis of Navier–Stokes solutions. The point is only that linear growth of a history clock does not by itself indicate finite-time blowup. Nothing here proves or disproves Fefferman Statements (C) or (D).

1 Scope, Scientific Honesty, and Non-Claim Boundary

Worked illustration of why a diverging quadratic history clock on $[0, \infty)$ is not a finite-time BKM blowup proof; explicitly declines to certify Fefferman (C) or (D).

- **Repository:** `https://github.com/BenFrohman/Clay-CD-not-certified`
- **Zenodo DOI (v1.0.0):** `10.5281/zenodo.23653015`
- **Earliest Git Commit SHA:** `bf14a32b0944c88da889ce99ceaee2901df53802` (2026-10-07 07:27:11 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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This repository is not a resolution of Fefferman (C) or (D).

A request to title a repository as a Clay C/D resolution, and to certify a zero-sorry blowup proof from the Section 10 argument, is declined. The Section 10 materials do not prove that the Beale–Kato–Majda integral diverges in finite time.

Blowup in the BKM sense is divergence of $\int \|\omega\|_\infty \, dt$. In the 2D and steady-Euler example M is bounded, so $\int_0^T M \, dt = MT$ stays finite while $\int_0^T M^2 \, dt$ grows. That is not a blowup proof.

Forced (C) and (D) are a different statement. An external claim exists in openai/NavierStokesAndEuler. This repository does not copy it and does not certify it.

3 Build and axiom status

Proved (kernel-checked). `lake build` of the `NotBlowup` library (Lean v4.30.0-rc1, Mathlib) succeeds. The axiom report (`lean/AxiomReport.lean`, output in `AXIOMS.md`) lists 2 theorems, `bounded_M_keeps_bkm_finite` and `history_can_grow_while_bkm_stays_finite`, with no `sorryAx` and no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. Both are definitional identities closed by `rfl`; they are illustrations, not mathematical results, and this record contains no mathematical theorem about Navier–Stokes.

Conditional on. Nothing; there are no hypotheses.

Open. Fefferman Statements (B), (C) and (D). This repository says nothing about them beyond the illustration above.

4 Provenance hashes

File fingerprints (`provenance/PROVENANCE_HASHES.json`, Merkle root in `provenance/PROVENANCE_MERKLE_RO`
Ed25519 key fingerprint `b428fb72676f1c52`) cover an earlier snapshot. They are integrity fingerprints only, not a certificate of correctness or review.

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